Saving, Investment and the Financial System

Mbak 9515: Economics

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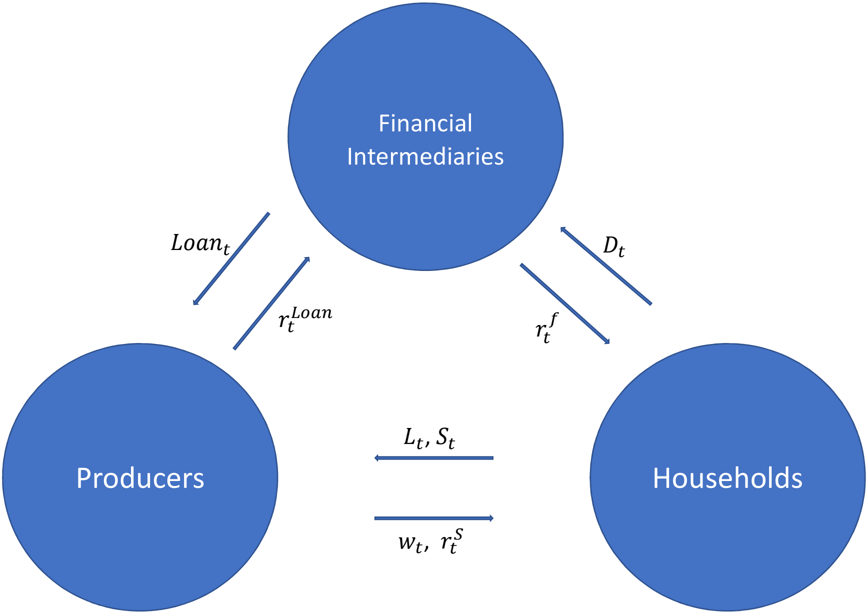
Readings

- Chapter 8. Saving, Investment, and the Financial System
  - Financial Markets
  - Accounting Identity
  - The Market for Loanable Funds

Basic Facts about Financial Structure



Financial Institutions

- **Financial system**
  - Group of institutions in the economy
  - That help match one person's saving with another person's investment
  - Moves the economy's scarce resources from savers to borrowers

- **Financial institutions**
  - Financial markets
  - Financial intermediaries

Financial Markets

- Savers can directly provide funds to borrowers
  - The bond market
  - The stock market

Financial Markets

- The bond market
  - Bond: certificate of indebtedness
    - Date of maturity, when the loan will be repaid
    - Rate of interest, paid periodically until the date of maturity
    - Principal, amount borrowed
  - Borrowing from the public
    - Used by large corporations, the federal government, or state and local governments

Financial Markets

- The stock market
  - Stock: claim to partial ownership in a firm
    - A claim to the profits that a firm makes
  - Organized stock exchanges
    - Stock prices: demand and supply
  - Equity finance
    - Sale of stock to raise money
  - Stock index
    - Average of a group of stock prices

Financial Intermediaries

- **Financial intermediaries**
  - Savers can indirectly provide funds to borrowers
  - Banks
  - Mutual funds

Financial Intermediaries

- **Banks**
  - Take in deposits from savers
    - Banks pay interest
  - Make loans to borrowers
    - Banks charge interest
  - Facilitate purchasing of goods and services
    - Checks: medium of exchange

Financial Intermediaries

- **Mutual funds**
  - Institution that sells shares to the public
  - Uses the proceeds to buy a portfolio of stocks and bonds
  - Advantages: diversification; professional money managers

National Saving and Its Components

- The portion of income not spent on consumption but spent on acquiring wealth by *households, firms and governments* will constitute the national saving

- Recall national income identity:

$$Y = C + I + G + NX$$

- For simplicity, let's assume that $NX = 0$:

$$Y = C + I + G$$

- Recall the definition of saving:
  - current income minus spending on *current needs*

National Saving and Its Components

- Consider investment

$$Y = C + I + G$$

- The acquisition of new factories, equipment, other capital goods, and residential construction
  - Investment spending is for the economy's *future* productive capacity
  - *not* for the current needs

- In contrast, we treat consumption and government spending as spending on *current* needs

National Saving and Its Components

- National saving: the saving of the entire economy

$$S = Y - C - G$$

- Private and public saving

$$S = Y - C - G + T - T$$

  - **transfer payments** (T, tax): payments the government makes to the public for which it receives no current goods or services in return
  - private saving (S_{private}): the saving of the private sector of the economy
  - public saving (S_{public}): the saving of the government sector

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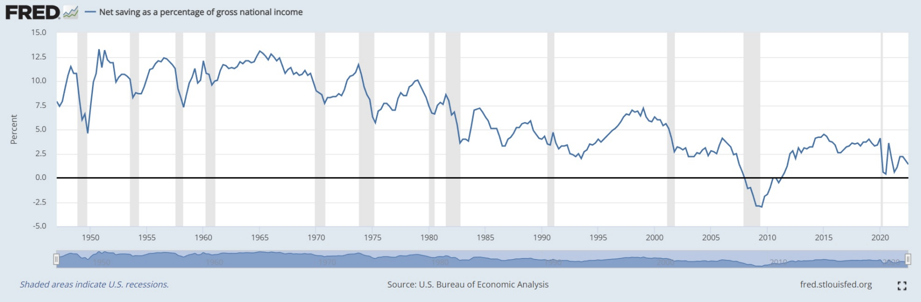

Government Budget

Public Saving

- Government budget **deficit**
  - the excess of government spending over tax collections
  - $T < G$ or $T - G < 0$
- Government budget **surplus**
  - the excess of government tax collections over government spending
  - $T > G$ or $T - G > 0$

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Government Budget

Example

Federal Government	
Receipts	3,452.1
Expenditures	4,149.4
Local Governments	
Receipts	1,303.1
Expenditures	1,293.2

- Find the federal government's budget surplus or deficit, local governments' budget surplus or deficit, and the contribution of the government sector to national saving
  - Federal government: 3,452.1-4,149.4=-697.3
  - Local government: 1,303.1-1,293.2=9.9
  - public saving: -697.3+9.9=-687.4

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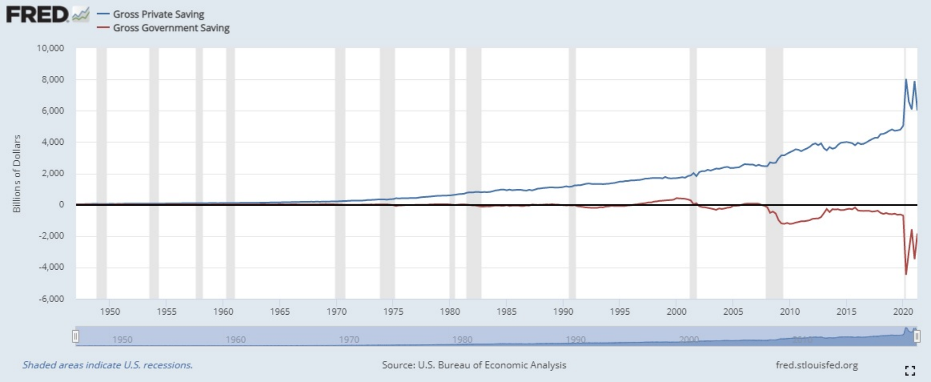

Investment and Capital Formation

- Why do we care saving?
  - Saving $\Rightarrow$ Investment $\Rightarrow$ Capital
  - Recall $S = Y - C - G = I$
  - Investment is the creation of new capital

- When do firms decide to invest?
  - revenue $>$ cost of investment

Investment and Capital Formation

Example

- Larry is thinking of going into the lawn care business. He can buy a \$4,000 riding mower by taking out a loan at 6% annual interest. With this mower and his own labor, Larry can net \$6,000 per summer, after deduction of costs such as gasoline and maintenance. Of the \$6,000 net revenues, 20% must be paid to the government in taxes. Assume that Larry could earn \$4,400 after taxes by working in an alternative job. Assume also that the lawn mower can always be resold for its original purchase price of \$4,000. Should Larry buy the lawn mower?

- We need to compare total revenue and cost
  - revenue after tax: \$6,000 $-0.2\times$\$6,000=\$4,800
  - cost: interest payment $0.06\times$\$4,000=\$240
  - Larry can earn \$4,800-\$240=\$4,560 > \$4,400

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- If the loan rate is 12%, not 6%?
  - revenue after tax: \$6,000 -0.2×\$6,000=\$4,800
  - cost: interest payment 0.12×\$4,000=\$480
  - Larry can earn \$4,800-\$480=\$4,320 < \$4,400
- If the price of mower is \$7,000, not \$4,000?
  - revenue after tax: \$6,000 -0.2×\$6,000=\$4,800
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Investment and Capital Formation

Factors that Affect Investment

- A decline in the price of new capital goods
- A decline in the real interest rate
- Technological improvement that raises the marginal product of capital
- Lower taxes on the revenues generated by capital
- A higher relative price for the firm's output

Investment and Capital Formation

Example 2

- Economists in Funlandia, a closed economy, have collected the following information about the economy for a particular year:

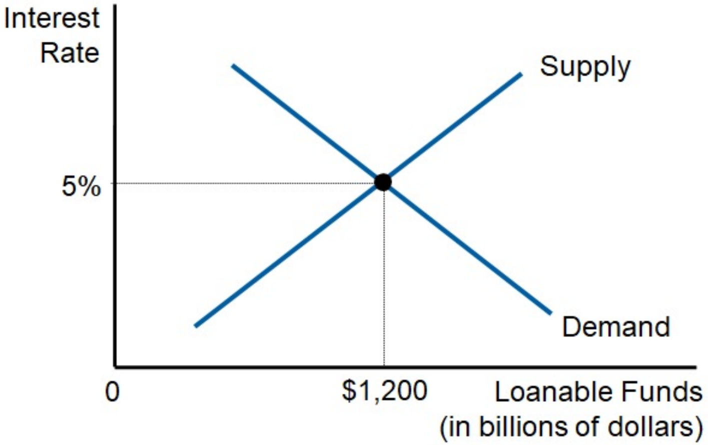
$$Y = 10,000$$
$$C = 6,000$$
$$T = 1,500$$
$$G = 1,700$$
$$I = 3,300 - 100r$$

  - Calculate private saving, public saving, national saving, invetment, and the equilibrium real interest rate.

The Market for Loanable Funds

- We saw that $S = Y - C - G = I$
  - Saving is the same as investment if $NX = 0$
- Who supplies saving?
  - Households, firms and the government
  - Upward-sloping saving curve
- Who demands savings?
  - firms that want to purchase or construct new capital (=investment)
  - Downward-sloping investment curve

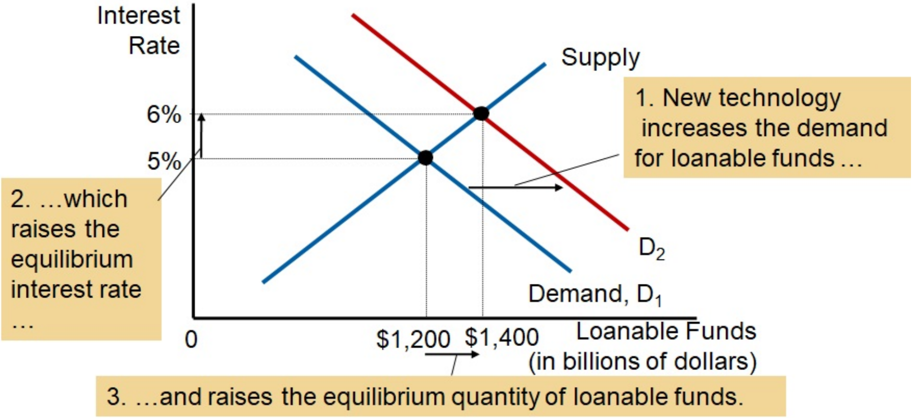
The Market for Loanable Funds



Saving, Investment and Financial Markets

Forces Affect the Demand Curve

- New technology: increases profit opportunities



Saving, Investment and Financial Markets

Forces Affect the Supply Curve

- Expansionary fiscal policy
  - The government increases its spending without raising taxes
  - This increases government budget deficit